

About Me

Name – Rushi Prajapati

Company Designation - AI Technology - Delivery & Support Engineer @ Sahana System Limited

Executive Summary

An avid learner, researcher, and community builder with a profound passion for Artificial Intelligence, dedicated to leveraging technology to tackle real-world challenges. Committed to sharing knowledge and fostering collaboration through talks and blogs within the community.

Education

University Name - Silver Oak University

College Name – Silver Oak College of Engineering and Technology

Course Name - Bachelor of Technology in Computer Engineering with Specialization of Artificial Intelligence and Machine Learning

College location - Ahmedabad, India

College Education Tenure - August 2020 – August 2024

CGPA: 9.87/10

Key Coursework:

Natural Language Processing, Machine Learning, Artificial Intelligence, Deep Learning and Reinforcement Learning, Computer Vision, Data Structures & Algorithms, Computer Networks, Cloud Computing, Software Engineering, Object Oriented Programming, Python programming

Achievements:

Ranked 22nd out of 1,213 students in the Bachelor of Technology program.

Achieved 3rd position out of 44 students in the specialization batch of Artificial Intelligence and Machine Learning.

This educational journey has provided a solid foundation in both theoretical knowledge and practical skills, preparing me to tackle contemporary challenges in technology and contribute effectively to the evolving field of AI.

Professional Experience

1. Full Time Work:

AI Technology – Delivery & Support Engineer, Sahana System Limited

June 2024 – Present

Ahmedabad, India

Multimodal Interactive Assistant Development: Engaged in creating interactive assistants utilizing Text-to-Speech (TTS) and Speech-to-Text (SST) engines, along with face-lip-sync models based on Generative Adversarial Networks (GANs).

Project Management: Responsible for preparing professional project proposals, designing intricate system architectures, and cultivating productive client relationships.

NLP Solutions: Developing document comparison and summarization tools using advanced Natural Language Processing (NLP) frameworks.

2. Internship:

AI Technology – Delivery & Support Intern, Sahana System Limited
December 2023 – May 2024
Ahmedabad, India

Object Detection & Tracking Models: Developed various models for object detection and tracking, leveraging Convolutional Neural Network (CNN) architectures and pretrained models to align with business logic and downstream applications.

Chatbot Development: Built a Retrieval-Augmented Generation (RAG) based offline chatbot using the LangChain framework and various open-source Large Language Models (LLMs), resulting in a 25% improvement in response accuracy and a 30% reduction in hallucinations.

Position of Responsibility

Leadership Roles:

1. Chairperson, IEEE Special Interest Group for Humanitarian Technology (SIGHT) Student Branch Group
Tenure - January 2023 – January 2024
2. Section Representative, IEEE SIGHT Gujarat Section
Tenure - January 2023 – January 2024
3. Vice-Chairperson, IEEE Women in Engineering (WIE) Affinity Group
Tenure - January 2023 – January 2024
4. Founder & Lead, IEEE Silver Oak University ML Geeks Club
Tenure - January 2023 – Present

Research Experience

1. Chapter Contribution:

Title: Reinforcement Learning-Based Data Dissemination

Publication: Springer

Expected Release: November 2024

Description: This chapter will be featured in the book Deep Learning Based Solutions for Vehicular Ad-hoc Networks, focusing on innovative methodologies for data dissemination in vehicular networks using reinforcement learning techniques.

2. Research Paper:

Title: Enhancing Image Understanding in Automatic Captioning

Publication: Elsevier

Expected Release: December 2024

Description: This paper explores the application of Spatially-Aware Transformer Architectures to improve automatic image captioning. The research aims to enhance the model's ability to generate coherent and contextually relevant descriptions by leveraging advanced deep learning techniques.

3.

Volunteer ship

1. Polygon Technology:
Position - Volunteer
Tenure - November 2022

Volunteered at a Polygon/Guild roadshow in Ahmedabad, contributing to the success of Gujarat's largest web3 conference. The event hosted over 300 attendees and featured 15 speakers from across India, creating lasting memories and valuable connections within the tech community.

2. AWS Cloud Club at Silver Oak University:
Position - Member & Volunteer
Tenure - December 2022 – January 2024

As an active member of the AWS Cloud Club, I regularly attended meetings and events hosted by the group. In addition to participating in these events, I volunteered my time to help organize and run AWS Cloud Club activities. This involvement provided opportunities to collaborate with like-minded individuals who share a passion for cloud computing and AWS technologies. Through these interactions, I gained insights into the latest trends and innovations in the field while sharing my own knowledge and experiences.

3. Google Developer Student Club (GDSC) at Silver Oak University:
Position - Member & Volunteer
Tenure - August 2023 – September 2024

As a dedicated member of GDSC SOU, I actively participate in organizing and executing a wide range of tech events. This involvement has allowed me to gain valuable experience in event management, including planning, coordination, and execution. An avid learner, I continuously seek opportunities to enhance my skills and knowledge in the ever-evolving tech landscape.

Journey

Growing up fascinated by science fiction depictions of AI in science fiction, like Iron Man's JARVIS, I found myself captivated by the potential of technology to solve real-world problems. This early inspiration sparked a journey of discovery that has profoundly shaped who I am today and fueled my passion for graduate study in AI-ML. Whether it is programming basic automation scripts or experimenting with AI-driven applications during my formative years, my curiosity has always been rooted in understanding the intricate logic behind complex algorithms and architectures. Nurtured by the unwavering support of my parents and mentors, as well as my profound fascination with Artificial Intelligence, pursuing a degree in Bachelors of Computer Engineering with a specialization in AI-ML emerged as the inevitable path for me. My undergraduate journey was a balance of theory and practicals. Disciplines like NLP, Deep Learning, Computer Vision, and Reinforcement Learning intrigued me to explore beyond the curriculum. This passion deepened my understanding of AI technologies and fueled my academic success, resulting in a 9.87 CGPA and a top 2% ranking in my department.

As part of my academic journey, I undertook projects that rigorously tested my AI knowledge and practical skills. Focusing on AI for public safety, I developed a computer vision-based casualty (falling person) detection system for public spaces, earning recognition from faculty, peers, and professionals. Encouraged by this success, I further tackled AI-driven substance abuse detection in public areas and schools, as well as animal and child abuse detection using advanced architectures and Transfer Learning. For the university, I created a Generative AI-powered interactive kiosk machine, merging advanced Text-to-Speech and Speech-to-Text engines with GANs for face-lip-syncing. I marveled at how this combination could create immersive user experiences. These projects, alongside my open-source AI contributions, honed my technical expertise and solidified my commitment to applying AI for societal benefit.

In my final year of undergraduate study and in the initial stage of my professional life, my fascination with AI, particularly reinforcement learning (RL), began to crystallize. Motivated by the potential of RL to dynamically adapt to complex, ever-changing environments, I dove deeper into exploring how this technology works. This interest led me to co-author a chapter in the upcoming Springer book "Deep Learning Based Solutions for Vehicular Ad-hoc Networks," expected in November 2024. The chapter, "Reinforcement Learning-Based Data Dissemination for Vehicular Ad Hoc Networks," delves into how RL can optimize data dissemination in VANETs, a critical element for intelligent transportation systems. My research tackles the dynamic nature of these networks, where conventional methods struggle with high node mobility and scalability, offering a robust RL-based solution that enhances network performance in real time. Parallel to this, my passion for AI-driven image understanding inspired another key research project focused on automatic image captioning. Co-authoring a paper titled "Enhancing Image Understanding in Automatic Captioning using Spatially-Aware Transformer Architectures," slated for publication in Elsevier by December 2024, I explored how transformer models could bridge the gap between image semantics and text generation. Also I'm engaged in a very interesting research topic—using RL in cloud load balancing. This avenue of research continues to fuel my passion for pushing the boundaries of AI applications in diverse, impactful areas.

I've always believed that the best way to truly understand something is to teach it, and this belief shaped much of my journey in the AI community. During my time at university, I founded ML Geeks, a club that quickly grew to over 100-120 students eager to explore data science, cloud computing, and AI-ML. I found great joy in simplifying complex topics like Data Science, Cloud 101, and ML-AI, and helping others grasp these challenging concepts. Through the club, I led numerous workshops and talks, not only sharing knowledge but also fostering curiosity and confidence among the students. A standout moment was delivering a talk at R10's Outstanding Student Branch of 2021, IEEE SOU SB which has many more significant awards, where I had the opportunity to engage with a wider community. These experiences solidified my love for teaching and deepened my commitment to contributing to the AI field through education and mentorship.

During my internship at Sahana System Limited, under the guidance of my professional mentor, I and my team designed an Automatic AI-powered Parade evaluation system using posture recognition and analysis at Sahana System Limited, where we used real time footage of parade ground and gave instant feedback on their mistake and synchronization check for contingent. Each step of the process was a thrilling exploration of AI Applications in the real world. Additionally, I delved into the intricacies of developing NLP-based document comparison and summarization tools, where each challenge became an opportunity for growth. These experiences, rich with learning and discovery, reinforced my commitment to harnessing AI for meaningful societal impact. These projects equipped me with invaluable industry experience, bridging theory with real-world AI challenges.

My long-term career goal is to leverage advanced AI techniques to mimic human cognitive skills by achieving Artificial General Intelligence, positioning myself as a leader in the AI field. My focus will be on transformative areas such as Brain-Computer Interfaces (BCI), Human-Computer Interaction (HCI), and autonomous systems. Through this specialization, I aim to tackle complex challenges through AGI that only a select few highly intelligent individuals can address, driving innovation and making a meaningful impact on society.

Blogs

1. Simplifying the World of Computer Vision

Description - Have you ever wondered how your smartphone recognizes faces or objects in photos? Welcome to the world of Computer Vision! In this engaging blog, I simplify key concepts such as image processing techniques and object recognition algorithms. Through relatable examples and practical applications, you'll discover how computer vision is revolutionizing technology—from security systems to self-driving cars. Join me as we explore this dynamic field that bridges technology and visual perception.

Blog Link: <https://rushiprajapati.medium.com/simplifying-the-world-of-computer-vision-6fb505b8ac9b>

Blog Image: <https://drive.google.com/file/d/1GHnZCoyhnX1hx-vzrP8x8nhZa1mkuHQD/preview>

2. Insights into Generative AI

Description - Generative AI is reshaping creativity by enabling machines to produce original content—from art to music. In this blog, I dive into what generative models are all about and their implications across various domains. By examining both their creative potential and ethical challenges, I invite you to join me on a thought-provoking journey that challenges traditional notions of creativity and authorship in the age of AI.

Blog Link: <https://rushiprajapati.medium.com/insights-of-generative-ai-8638ce0b1017>

Blog Image: <https://drive.google.com/file/d/13mc-ggHYaZBteY4Pkf5DziDcc35JbPeY/preview>

3. Simplifying the World of Machine Learning & Deep Learning

Description - Machine Learning and Deep Learning are transforming industries at an unprecedented pace—but what do they really mean? In this comprehensive blog, I break down these complex fields into understandable segments. By exploring key differences and practical applications, I aim to demystify these technologies for readers at all levels. Join me as we navigate through algorithms and use cases that showcase the incredible potential of ML and DL in shaping our future.

Blog Link: <https://rushiprajapati.medium.com/simplifying-the-world-of-machine-learning-deep-learning-9f05f0989274>

Blog Image: <https://drive.google.com/file/d/1VuhCVdw7A2FVdGngpB5FZQ2Tw-d0KNwC/preview>

4. The ABCs of Neural Networks

Description - Are you curious about how machines learn from data? In 'The ABCs of Neural Networks,' I provide an engaging introduction to the fundamental concepts that underpin this revolutionary technology. From neurons to training processes, I simplify complex ideas into relatable terms that anyone can understand. This blog serves as your gateway into the world of neural networks, paving the way for further exploration into machine learning's vast landscape.

Blog Link: <https://rushiprajapati.medium.com/the-abcs-of-neural-networks-2ad111779c93>

Blog Image: <https://drive.google.com/file/d/13GwU0fPKM26dVLeGU7-P8TTwtoF5byNv/preview>

5. Demystifying the Decision Makers of Neural Networks

Description - What goes on inside a neural network when it makes decisions? In this blog, I take you behind the scenes to explore the critical components that drive neural networks' decision-making processes. By breaking down concepts like activation functions and layers into digestible parts, I aim to provide clarity on how these systems learn from data. Whether you're new to deep learning or seeking to deepen your understanding, this exploration promises valuable insights into one of AI's most intriguing aspects.

Blog Link: <https://rushiprajapati.medium.com/demystifying-the-decision-makers-of-neural-networks-dcfb536d915a>

Blog Image:
<https://drive.google.com/file/d/1Hcy4b7SRAv081JBKfjNj8GAMIT6jrOkj/preview>

6. Data: The New Oil

Description - Data has become the lifeblood of innovation in our digital age—akin to oil during the industrial revolution. In this thought-provoking blog, I explore why data is often referred to as 'the new oil' and discuss its implications for businesses and society at large. From ethical considerations to success stories that highlight data's transformative power, this post invites you to rethink your relationship with data and consider its potential as a strategic asset in your endeavors.

Blog Link: <https://rushiprajapati.medium.com/data-the-new-oil-d62cfb673c35>

Blog Image:

https://drive.google.com/file/d/1Y8scpmYjxaCXLlZobrdXDHew0nVDDU_g/preview

7. Understanding Data

Description - In today's data-driven world, understanding how to harness the power of data is crucial for success. In this blog, I delve into the significance of data quality, preprocessing techniques, and how informed decision-making can transform businesses. Through practical insights and real-world scenarios, I aim to illuminate the path to effective data management. Whether you're a business leader or an aspiring data scientist, this exploration will empower you to leverage data as your most valuable asset.

Blog Link: <https://rushiprajapati.medium.com/understanding-data-109e0d7eb795>

Blog Image:

https://drive.google.com/file/d/1kAyKvl_jFhdehh3eN7hW9uQRNXbLRrX8/preview

8. CNN in a Nutshell

Description - What if machines could see and interpret images just like humans? Enter Convolutional Neural Networks (CNNs), the technology behind computer vision. In this blog, I simplify the complexities of CNNs, explaining how they recognize patterns and objects within images. With relatable examples and clear explanations, you'll gain an appreciation for how CNNs are revolutionizing industries—from healthcare to autonomous vehicles. Join me on this visual journey that demystifies one of the most exciting advancements in artificial intelligence.

Blog Link: <https://rushiprajapati.medium.com/cnn-in-nutshell-2959bdcbb3c1>

Blog Image:

https://drive.google.com/file/d/1BkO0sc_tNSbM3hDTMtQfJRuRarPJ3LLC/preview

9. Learning to Win: Reinforcement Learning Demystified

Description - Imagine teaching a computer to play a game by rewarding it for good moves and punishing it for bad ones. This is the essence of Reinforcement Learning (RL), a fascinating area of machine learning that mimics how we learn from our experiences. In this blog, I break down the core concepts of RL, making them accessible and engaging. From

understanding reward systems to exploring real-world applications, this post will equip you with a solid foundation in RL, inspiring you to delve deeper into its transformative potential.

Blog Link: <https://rushiprajapati.medium.com/learning-to-win-reinforcement-learning-demystified-5ac2a14a0fe8>

Blog Image:

https://drive.google.com/file/d/18Us2XNb3_wS5YDhSHTUVln6wPDaOsRlb/preview

10. Demystifying Dynamic Programming and Linear Programming from the Perspective of AI and Reinforcement Learning

Description - Have you ever wondered how AI systems make complex decisions or optimize processes in real-time? In this blog, I unravel the intricate worlds of Dynamic Programming (DP) and Linear Programming (LP), two powerful techniques that are the backbone of efficient algorithms in artificial intelligence. Join me as we explore how these methodologies not only solve optimization problems but also enhance decision-making in Reinforcement Learning.

Blog Link: <https://rushiprajapati.medium.com/demystifying-dynamic-programming-and-linear-programming-from-the-perspective-of-ai-and-15c9d0af13ac>

Blog Image: <https://drive.google.com/file/d/1-4Z35-hHYrsVnYHSlgsHvfdGfCYSIxp/preview>

Talks

1. EDUTALK: CLOUD 101

Date: 20th April, 2023

2. GEEK OUT: CHATBOTS AND LLM IN A DAY

Date: 7th March, 2024

3. PARAM Shavak: ML Unleashed

Date: 23rd March, 2024

4. EDU-TALK: Fundamentals Of Machine Learning

Date: 4th July, 2022

5. IEEE SIGHT Enlightenment Drive

Date: 8th May, 2023

Talks with Description:

1. EDUTALK: CLOUD 101

Date: 20th April, 2023

I had the privilege of conducting the Edu-Talk 'Cloud 101: Get Ahead In Cloud'. The session aimed to provide attendees with a comprehensive overview of cloud computing and its significance, along with practical guidance on deploying a website using Amazon Web Services (AWS). Through engaging discussions and a live demonstration, participants gained insights into the transformative potential of cloud technology. The interactive format encouraged curiosity and exploration, leaving attendees inspired to delve deeper into the world of cloud computing.

Talk Report Link: https://drive.google.com/file/d/1MT6urWxV8pcCFFc_jgWCycm2ke-cxS7C/view?usp=sharing

Talk Photos:

1. <https://drive.google.com/file/d/1DtqIFuBmHSdX97TJq3umXQeAwxbmo4u-/preview>
2. <https://drive.google.com/file/d/15kTrMlJMVmKKhFKaoX-1JuckDCu2dCOR/preview>
3. https://drive.google.com/file/d/1k4IwI0-9s7L9aBAMIsPZ_HqlsSGHh2r5/preview
4. <https://drive.google.com/file/d/1suLnyKIYx6CLAAInTHKt21qWe1MJaLSF/preview>
5. https://drive.google.com/file/d/1QNTSFo1LxpHZz828NZB8obXs-90nz_eo/preview
6. <https://drive.google.com/file/d/1puJrBeC5r8vdNx4KNXQ0EomlfwF6Mb6B/preview>

Link for Each Photos:

https://drive.google.com/drive/folders/1OUFdRoVN5DE4NkAmvYxQwZeftA_wlf70?usp=sharing

2. GEEK OUT: CHATBOTS AND LLM IN A DAY

Date: 7th March, 2024

'Geek Out: Chatbots and LLM in a Day' session/event explored the intricacies of chatbots and Large Language Models (LLMs), providing participants with a comprehensive understanding of these cutting-edge technologies. The session featured engaging discussions on foundational concepts such as artificial intelligence, machine learning, and deep learning, alongside practical applications in natural language processing. Participants enjoyed a hands-on technical session where they developed models using the university's supercomputer 'PARAM Shavak'.

Talk Report Link:

<https://drive.google.com/file/d/12Pv3zuNHSBq8KdXdrOz3Ug3R4XMieKSZ/view?usp=sharing>

Talk Photos:

1. <https://drive.google.com/file/d/1CORgb2hladyDGT9qMb9UWbdPue6CCJF/preview>
2. <https://drive.google.com/file/d/1fEFHvI4fuxvn2JoCV6kkMuIsDazq5xxQ/preview>
https://drive.google.com/file/d/1LW1a9CqP8WHL4-0Kj9tloVL_YxCMImuY/preview
3. https://drive.google.com/file/d/1peviaDiMIMOR78_ju8NIO5a6_f8hb1b1/preview
4. https://drive.google.com/file/d/1j5WPYtZwZ6iwmk6oarUaGMc_z4RqA1xf/preview
5. <https://drive.google.com/file/d/1tjvfvZ1vEOSB4spDPIDa-34yjNpQlyq0/preview>
6. https://drive.google.com/file/d/1VjOLNs_PuO8LpbbUJkaucEmuNAV1i-i9/preview
7. https://drive.google.com/file/d/1IQYUINP_h0jxfPwOg6Afd-6nqYr2Cy6S/preview
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Link for Each Photos:

<https://drive.google.com/drive/folders/1TLgquuZfdRxPANTG2btZhqmd5-g50hDr?usp=sharing>

3. PARAM Shavak: ML Unleashed

Date: 23rd March, 2024

I co-hosted the event 'PARAM Shavak: ML Unleashed'. This workshop focused on two critical areas: machine learning model optimization and Responsible AI practices. Alongside my fellow club leads, I engaged participants in discussions about the intricacies of machine learning, including the optimization of models and the ethical considerations necessary for developing responsible AI systems.

Talk Report Link: https://drive.google.com/file/d/1VbHuLKzXcVO6mjX-FDfoh26_0bQx9cLJ/view?usp=sharing

Talk Photos:

1. <https://drive.google.com/file/d/1uWRktKWzb0XRwC2OTbb6H0kPmSGIATtR/preview>
2. <https://drive.google.com/file/d/1BxivZ9ysrWmaeMrt5rlSGS0hJA4ihDOA/preview>
3. <https://drive.google.com/file/d/1hF7WCfcam3puCywS4w7Q6bPnglVDMbZQ/preview>
4. https://drive.google.com/file/d/1N15evR5jhsXR4jP_pXXui8UiseRI9Xjr/preview
5. <https://drive.google.com/file/d/1HCgUxxbo1JFr3TUG1ohqOKtme1f7ih2H/preview>
6. https://drive.google.com/file/d/132wlg7138Vgbd9JpipadHNJUnpO4H_tj/preview

7. https://drive.google.com/file/d/1fAtX6UY6H_e72PW6ToEddA6vzqin8Dg/preview
8. <https://drive.google.com/file/d/14d6wRDFEWQZ3o6yZjWhEkQ5eiYHDGzz/preview>

Link for Each Photos:

<https://drive.google.com/drive/folders/1xdgmIBxZG9ER4ekyCsQmlHx1yeHn8Wur?usp=sharing>

4. EDU-TALK: Fundamentals Of Machine Learning

Date: 4th July, 2022

I conducted the Edu-Talk titled 'Fundamentals of Machine Learning'. The session aimed to provide participants with a comprehensive understanding of machine learning, including its history, use cases, and comparisons between human brains and machines. The event was well-received, culminating in enthusiastic applause for the insights shared throughout the talk.

Talk Report Link:

<https://drive.google.com/file/d/1rm5ewCl8iDBWs2FQXhMGeEA3uk4oucxB/view?usp=sharing>

Talk Photos:

1. <https://drive.google.com/file/d/1d0XCWMM6byGVkOL9HWkFY2hRkYc1dOwr/preview>
2. <https://drive.google.com/file/d/1Rr9Zqh1S6pqgdrCs2STUNwjA8qeBLCfB/preview>

Link for Each Photos:

https://drive.google.com/drive/folders/1WPN6mfwdYThmmpGn_jJvYIsMkxYH1zUi?usp=sharing

5. IEEE SIGHT Enlightenment Drive

Date: 8th May, 2023

As Chairperson of the IEEE Special Interest Group for Humanitarian Technology (SIGHT) SOU Student Branch Group, I delivered a talk in an event that successfully raised awareness about technology's transformative potential in addressing social and environmental challenges.

Talk Report Link: https://drive.google.com/file/d/1R_TkoxSxf52HEtqoR-WRLqwrbyLL1g9b/view?usp=sharing

Talk Photos: <https://drive.google.com/file/d/1ZtySoLbRYUVd-WmKcYNyqnevuMz1rvvo/preview>

Link for Each Photos:

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